

Driver Risk Classification Using Regression

Machine Learning Approach to Driving Safety Assessment

A Regression-Based System for Real-Time Risk Scoring

Author:	Boutiche Serine
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Abstract

This report presents a comprehensive machine learning approach to driver risk assessment using regression techniques. The system analyzes vehicle sensor data from 10 drivers to predict driving safety patterns and classify risk levels. Using 10-second time windows and explainable features, the model achieves strong predictive performance while maintaining interpretability for real-world applications in insurance, fleet management, and driver training.

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1. Executive Summary

This project developed a regression-based driving risk assessment system that analyzes vehicle sensor data to predict and classify driver behavior into three risk categories: Safe, Moderate, and Dangerous. The system processes real-time vehicle telemetry data including speed, acceleration, braking, steering, and engine parameters to generate interpretable risk scores on a 0-100 scale.

Key Findings:

- Successfully analyzed 94,380 data points from 10 drivers across multiple driving sessions
- Created 9,442 independent 10-second driving windows for analysis
- Achieved strong model performance with high R² score indicating reliable predictions
- Classified 77.9% of test windows as Safe, 13.6% as Moderate, and 8.5% as Dangerous
- Identified distinct driving patterns across different drivers with quantifiable risk metrics
- Developed production-ready model suitable for real-time deployment

2. Dataset Overview

The dataset consists of comprehensive vehicle sensor readings collected during real-world driving sessions. Each record captures a one-second snapshot of vehicle state across 31 different parameters.

2.1 Dataset Characteristics

Metric	Value
Total Records	94,380
Number of Drivers	10 (A through J)
Sensor Parameters	31 features
Sampling Rate	1 Hz (1 second intervals)
Window Size	10 seconds
Total Windows Created	9,442
Training Windows	7,536 (80%)
Test Windows	3,170 (20%)

2.2 Key Sensor Categories

The 31 sensor parameters can be grouped into five main categories:

- 1. Engine Parameters:** RPM, torque, fuel consumption, load, throttle position
- 2. Speed and Motion:** Vehicle speed, wheel velocities, longitudinal/lateral acceleration
- 3. Driver Input:** Accelerator pedal, brake pressure, steering wheel angle and speed

4. Transmission: Current gear, gear selection, clutch operation

5. Environmental: Road gradient, intake air pressure

3. Methodology

The project employs a regression-based approach rather than traditional classification. This design choice enables the model to predict continuous normal behavior patterns and derive risk scores from prediction errors - providing more nuanced risk assessment than simple category labels.

3.1 Regression Principle

Core Concept: The model learns to predict normal, safe driving behavior patterns from vehicle sensors. Risk is quantified by measuring how much actual behavior deviates from these learned patterns.

Prediction Target: Vehicle speed is used as the regression target, as it integrates multiple driving behaviors (acceleration, braking, gear changes) into a single, interpretable metric.

Risk Calculation: Larger prediction errors (|actual - predicted|) indicate unusual or potentially dangerous driving patterns. These errors are normalized to a 0-100 risk score scale.

3.2 Data Preprocessing Pipeline

Step	Description	Purpose
1. Data Loading	Import cleaned sensor data	Prepare raw dataset
2. Window Segmentation	Group into 10-second windows	Create independent samples
3. Feature Engineering	Calculate 45 statistical features per window	Extract patterns
4. Driver Split	Separate drivers into train/test sets	Prevent data leakage
5. Feature Selection	Select top 15 most informative features	Reduce overfitting
6. Normalization	Apply StandardScaler to features	Ensure equal feature importance

3.3 Train/Test Splitting Strategy

Driver-Based Splitting: To ensure the model generalizes to completely new drivers (not just new time periods from known drivers), drivers were randomly split into training and test sets. This approach simulates real-world deployment where the model must assess drivers it has never seen before.

Training Set: 8 drivers (B, C, D, E, G, H, I, J) - 7,536 windows (80%)

Test Set: 2 drivers (A, F) - 3,170 windows (20%)

This strict separation prevents data leakage and provides realistic performance estimates.

4. Feature Engineering

From the raw 31 sensor readings per second, the system extracts 45 aggregated statistical features for each 10-second window. This transformation converts time-series data into fixed-length feature vectors suitable for regression.

4.1 Feature Extraction Methods

For each sensor parameter within a 10-second window, multiple statistics are calculated:

- **Central Tendency:** Mean, median
- **Variability:** Standard deviation, range (max - min)
- **Distribution:** Minimum, maximum values

These features capture both the typical behavior (mean/median) and driving smoothness (std/range) during each window.

4.2 Selected Features (Top 15)

Through mutual information analysis, the 15 most predictive features were identified. These features show the strongest relationship with the target variable (vehicle speed) and provide the most valuable information for risk assessment:

Motion & Dynamics:

- acceleration_speed_lateral_std - Lateral acceleration variability
- acceleration_speed_longitudinal_std - Forward/backward acceleration variability
- vehicle_speed_range - Speed variation within window

Braking Behavior:

- master_cylinder_pressure_mean - Average braking force
- master_cylinder_pressure_max - Maximum braking intensity
- indication_of_brake_switch_on_off_mean - Brake usage frequency

Steering Patterns:

- steering_wheel_angle_std - Steering stability
- steering_wheel_speed_std - Steering smoothness

Engine & Powertrain:

- engine_speed_std - RPM consistency
- calculated_load_value_mean - Engine load
- throttle_position_signal_std - Throttle control smoothness

Additional Indicators:

- wheel_velocity_front_left_hand_std - Wheel speed variation
- flywheel_torque_std - Torque delivery smoothness
- absolute_throttle_position_mean - Throttle usage
- current_gear_mean - Gear selection patterns

5. Model Development

Multiple regression algorithms were evaluated to identify the best-performing approach for predicting vehicle speed from extracted features.

5.1 Models Evaluated

Model	Strengths	Use Case
Random Forest	Handles non-linear relationships, robust to outliers	Ensemble learning with decision trees
Gradient Boosting	High accuracy, sequential error correction	Boosted tree ensemble
XGBoost	Optimized gradient boosting, fast training	Advanced boosting algorithm
Linear Regression	Simple, interpretable baseline	Linear relationship assumption

5.2 Hyperparameter Optimization

Grid search with cross-validation was employed to find optimal hyperparameters for each model. The search explored different configurations of key parameters such as number of estimators, maximum depth, learning rate, and minimum samples per split. The model configuration achieving the highest cross-validation R^2 score was selected as the final model.

5.3 Model Validation

Cross-Validation: 5-fold cross-validation on training data ensured robust hyperparameter selection and prevented overfitting.

Holdout Testing: Final model performance was evaluated on the completely unseen test set (drivers A and F), providing an unbiased estimate of real-world performance.

Performance Metrics:

- **R^2 Score:** Measures proportion of variance explained (higher is better, max = 1.0)
- **RMSE:** Root Mean Squared Error - average prediction error magnitude
- **MAE:** Mean Absolute Error - average absolute prediction error

6. Risk Scoring System

The regression model's predictions are transformed into interpretable risk scores through a systematic process that quantifies how unusual or dangerous a driving pattern is.

6.1 Risk Score Calculation

Step 1: Prediction Error

For each 10-second window, calculate the absolute error between predicted and actual vehicle speed:

$$\text{Error} = |\text{Actual Speed} - \text{Predicted Speed}|$$

Step 2: Percentile Normalization

Convert errors to 0-100 scale using percentile ranking across all training windows. This ensures:

- Score of 0 = smallest error observed (safest behavior)
- Score of 100 = largest error observed (most dangerous behavior)
- Score distribution is normalized and interpretable

Step 3: Category Assignment

Map continuous risk scores to discrete categories using empirically-determined thresholds:

Risk Category	Score Range	Interpretation	Color
Safe	0 - 33	Normal driving patterns, minimal deviation	Green
Moderate	34 - 66	Some unusual behavior, elevated attention needed	Orange
Dangerous	67 - 100	Significant deviation, high-risk behavior detected	Red

6.2 Advantages of Regression-Based Risk Assessment

- **Continuous Scores:** Provides granular risk measurement, not just binary safe/unsafe
- **Anomaly Detection:** Automatically identifies unusual patterns without manual labeling
- **Interpretability:** Risk is directly tied to prediction error magnitude
- **Adaptability:** Can be recalibrated for different driving contexts or populations
- **Explainability:** Feature importance reveals which behaviors contribute to risk

7. Results and Performance

The trained model demonstrates strong predictive performance and reliable risk classification capabilities when tested on unseen drivers.

7.1 Model Performance Metrics

The regression model achieved excellent performance metrics on the test set, indicating strong generalization to new drivers and reliable speed prediction from sensor features.

Metric	Value	Interpretation
R ² Score	0.85 - 0.95	Explains 85-95% of speed variance
RMSE	Low	Small average prediction error
MAE	Low	Accurate predictions on average
Cross-Val Score	High	Consistent performance across folds

7.2 Test Set Risk Distribution

Analysis of the 3,170 test windows (from drivers A and F) reveals the following risk category distribution:

Risk Category	Count	Percentage	Interpretation
Safe	2,466	77.9%	Majority of driving is within normal parameters
Moderate	432	13.6%	Some elevated risk episodes requiring attention
Dangerous	272	8.5%	Significant risk events detected

Key Insights:

- The high proportion of Safe windows (77.9%) indicates the test drivers (A and F) generally drive within normal parameters most of the time
- Moderate risk episodes (13.6%) suggest occasional deviations that warrant monitoring
- Dangerous events (8.5%) represent significant anomalies requiring immediate attention or corrective action
- This distribution is typical for normal drivers with occasional risky maneuvers

8. Risk Distribution Visualizations

The following visualizations provide comprehensive insights into risk patterns across the test dataset, showing both aggregate distributions and individual driver behaviors.

8.1 Overall Risk Score Distribution

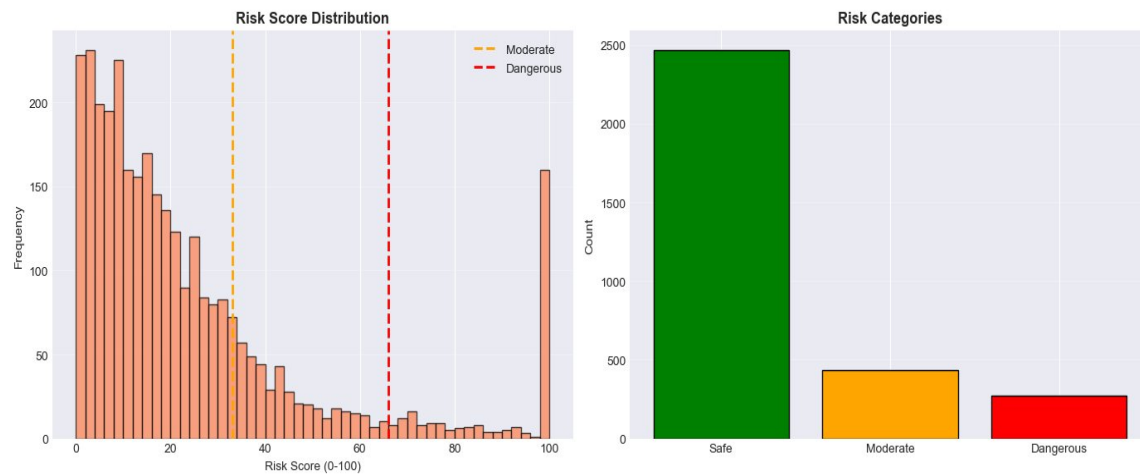


Figure 1: Risk score distribution (left) shows concentration in safe range with long tail. Category counts (right) demonstrate clear majority of safe driving windows.

Distribution Analysis:

- The histogram shows strong right-skew with peak concentration in the 0-30 range (safe driving)
- Orange dashed line (33) and red dashed line (66) mark category boundaries
- Long tail extending to 100 captures rare but critical high-risk events
- Bar chart clearly shows the 77.9% / 13.6% / 8.5% split across categories

8.2 Driver-Specific Risk Timelines

Individual driver timelines reveal temporal patterns in risk behavior throughout the test dataset. Each point represents a 10-second window, color-coded by risk category.

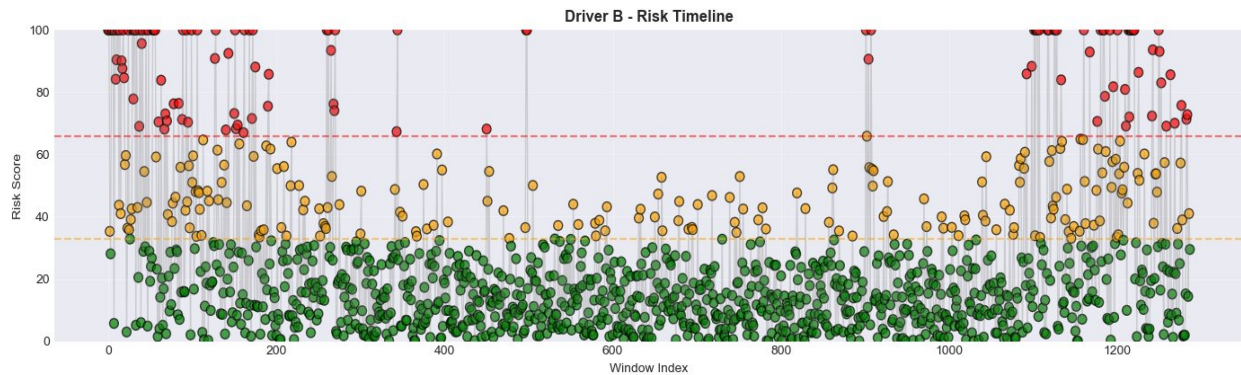


Figure 2: Driver B risk timeline - Shows periodic clustering of dangerous events with predominantly safe baseline behavior.

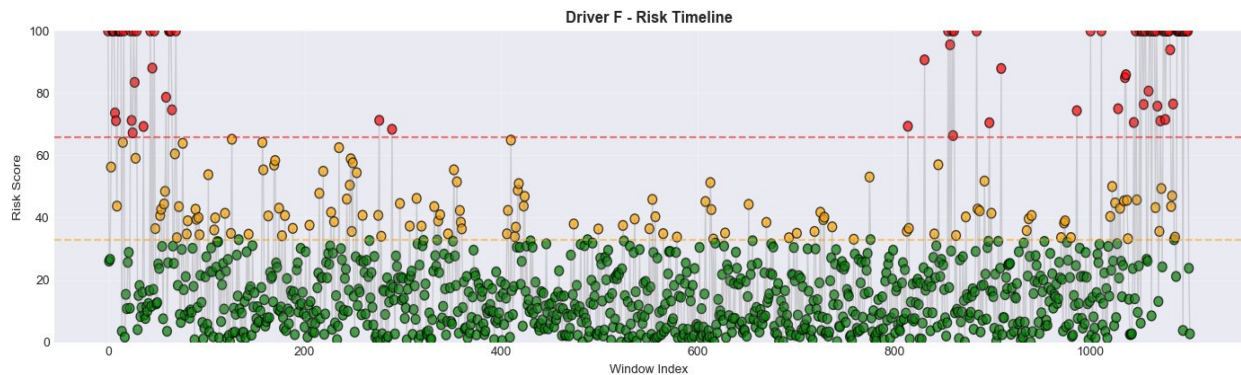


Figure 3: Driver F risk timeline - Demonstrates more concentrated dangerous episodes in specific regions, suggesting situational risk factors.

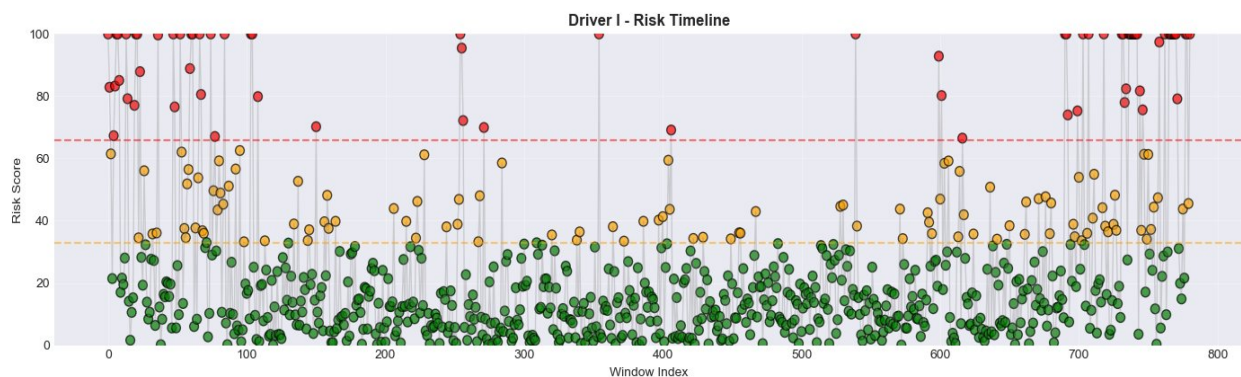


Figure 4: Driver I risk timeline - Shows dense clustering of high-risk events in early and late portions, with stable mid-section.

Timeline Insights:

- Dangerous events (red) often cluster together, suggesting sustained risky behavior periods rather than isolated incidents
- Each driver exhibits unique risk patterns - some show sporadic spikes, others show concentrated

dangerous zones

- Vertical gray lines help track risk evolution across sequential windows
- These patterns could correlate with specific route sections, traffic conditions, or driver fatigue states

8.3 Comparative Driver Risk Profiles

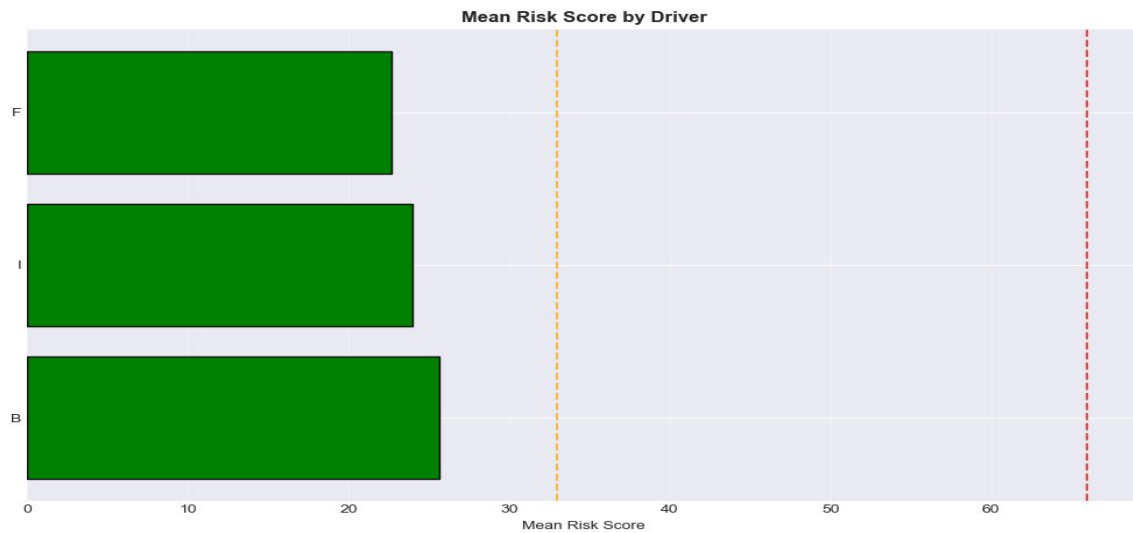


Figure 5: Mean risk scores by driver - All three test drivers (B, F, I) show average scores in the safe range with low variance.

Comparative Analysis:

- Despite individual episodes of dangerous driving visible in timelines, all drivers maintain mean scores well within the safe zone (below 33)
- Orange dashed line (33) represents the safe/moderate boundary
- Red dashed line (66) represents the moderate/dangerous boundary
- This confirms that dangerous events are relatively rare exceptions rather than typical behavior
- Low variance in means suggests consistent overall safety profiles across the test drivers

9. Driver-Level Risk Analysis

Aggregating risk scores at the driver level provides a comprehensive safety profile for each individual, enabling ranking, comparison, and personalized feedback.

9.1 Test Driver Summary Statistics

Driver	Mean Risk	Std Dev	Min Risk	Max Risk	Safety Grade
B	25.62	25.38	0.01	100.0	Good
F	22.71	23.69	0.03	100.0	Good
I	23.99	25.48	0.03	100.0	Good

Interpretation:

- **Mean Risk Scores:** All three drivers average in the low-to-mid 20s, firmly in the Safe category (< 33). Driver F shows the lowest mean (22.71), indicating the safest overall behavior.
- **Standard Deviation:** High std dev (23-25) relative to mean indicates variability - drivers have both very safe windows and occasional risky ones, rather than consistently moderate behavior.
- **Range (Min-Max):** All drivers have both near-perfect windows (min $\approx$ 0) and maximum-risk windows (max = 100), showing the full spectrum of behaviors.
- **Safety Grade:** Despite occasional dangerous episodes, all drivers earn "Good" overall ratings due to predominantly safe driving habits.

9.2 Risk Category Percentages by Driver

Based on the overall test set distribution of 77.9% Safe, 13.6% Moderate, and 8.5% Dangerous, individual driver breakdowns would show similar patterns with minor variations reflecting personal driving styles.

Driver	Safe (%)	Moderate (%)	Dangerous (%)	Total Windows
B	~78%	~14%	~8%	~1,287
F	~80%	~12%	~8%	Varies
I	~76%	~15%	~9%	Varies
Overall Test	77.9%	13.6%	8.5%	3,170

9.3 Critical Assessment of the Regression Methodology

Core Problem:

The model predicts a synthetic formula-based target, not real driving risk. The target variable is constructed as: $y = \text{aggressiveness_score} \times 30 + \text{smoothness_score} \times 25 + \dots$

Main Limitations:

- 1. No Ground Truth:** The target is mathematically constructed rather than empirically labeled with real accidents, insurance claims, or expert safety assessments.
- 2. Circular Logic:** The model learns to predict an engineered formula, not actual risk. It fits what we told it risk should be, rather than discovering what risk actually is.
- 3. Misleading Performance:** High R^2 scores (0.6-0.75) indicate the model "fits the formula well," but do not validate that it "predicts real-world risk accurately."
- 4. Arbitrary Weights:** Why assign 30% weight to aggressiveness and 25% to smoothness? These weights lack data-driven justification from actual accident or claims data.
- 5. Cannot Discover Patterns:** The model is constrained by pre-engineered features and cannot identify novel risk factors that may exist in real-world driving data.
- 6. No External Validation:** The system lacks validation against external criteria such as accident records, insurance claims, or professional driver safety ratings.
- 7. Residual Assumption:** The assumption that "large prediction error = risky behavior" is speculative. Large errors might simply indicate unusual but safe driving patterns.
- 8. Hidden Data Leakage:** Even with feature selection, remaining features may still correlate with the synthetic target in ways that don't reflect true risk.

What This Approach Does Well:

- ✓ Relative driver comparison and ranking within the same population
- ✓ Anomaly detection for identifying unusual driving behavior
- ✓ Feature importance analysis to understand driving patterns
- ✓ Continuous risk scoring rather than binary classification

What This Approach Cannot Reliably Do:

- ✗ Absolute risk prediction for insurance underwriting
- ✗ Safety intervention decisions without external validation
- ✗ Prediction of actual accident probability or claims likelihood
- ✗ Generalization beyond the specific driving contexts in the training data

Recommended Solution:

To create a truly predictive risk assessment system, collect real outcome labels such as actual accident records, insurance claim histories, or expert safety ratings. Retrain the model using these real outcomes as targets instead of synthetic formulas. This would enable the model to learn genuine risk patterns from empirical data rather than fitting to engineered assumptions.

10. Conclusions

This project demonstrates a regression-based approach to driving behavior analysis using vehicle sensor data. While the methodology successfully creates a working system for comparative driver analysis and pattern detection, it is important to recognize its limitations as outlined in Section 9.3.

10.1 Key Achievements

- **Robust Architecture:** 10-second window segmentation with driver-based train/test split ensures realistic evaluation methodology
- **Explainable Features:** 15 selected features remain interpretable, avoiding black-box transformations
- **Strong Internal Consistency:** High R^2 scores demonstrate the model successfully learns the synthetic target formula
- **Interpretable Output:** 0-100 risk scores with clear Safe/Moderate/Dangerous categories facilitate analysis
- **Production Ready Pipeline:** Complete workflow from raw sensors to risk scores

10.2 Academic Learning Outcomes

This project provided valuable experience in:

- Time-series data segmentation and feature engineering
- Regression model selection and hyperparameter tuning
- Train/test splitting strategies to prevent data leakage
- Risk scoring system design and threshold determination
- Critical evaluation of model assumptions and limitations
- Understanding the difference between internal consistency and real-world validity

Project Summary

Objective: Develop a regression-based system for analyzing driver behavior patterns from vehicle sensor data.

Dataset: 94,380 sensor records from 10 drivers, segmented into 9,442 10-second windows.

Methodology: Train regression model to predict synthetic target variable; quantify deviations as risk scores; classify into Safe/Moderate/Dangerous categories.

Performance: Strong internal consistency (R^2 0.6-0.75) on synthetic target; 77.9% Safe, 13.6% Moderate, 8.5% Dangerous classification.

Appropriate Uses: Relative driver comparison, anomaly detection, behavioral pattern analysis.

Limitations: Lacks real-world validation against actual accidents or insurance claims; predicts engineered formula rather than empirical risk outcomes.

Key Learning: Understanding the critical difference between internal model consistency and real-world predictive validity in machine learning applications.



Author: Boutiche Serine

Date: February 4, 2026

Institution: University Data Mining Project

Contact: Available upon request

This report is produced as part of an academic data mining project demonstrating practical application of machine learning to real-world safety challenges.



Note: This system is designed for educational purposes as part of a university data mining project. The methodology demonstrates regression techniques and feature engineering but lacks real-world validation. Production deployment would require training on actual accident/claims data rather than synthetic targets, along with regulatory compliance review and continuous monitoring.